

PERSONAL INFORMATION SYSTEM

PROGRAM

SET SERVEROUTPUT ON;

SET LINESIZE 120;

SET PAGESIZE 50;

BEGIN

EXECUTE IMMEDIATE 'DROP TABLE Employment CASCADE CONSTRAINTS';

EXCEPTION

WHEN OTHERS THEN NULL;

END;

/

BEGIN

EXECUTE IMMEDIATE 'DROP TABLE Education CASCADE CONSTRAINTS';

EXCEPTION

WHEN OTHERS THEN NULL;

END;

/

BEGIN

EXECUTE IMMEDIATE 'DROP TABLE Addresses CASCADE CONSTRAINTS';

EXCEPTION

WHEN OTHERS THEN NULL;

END;

/

BEGIN

EXECUTE IMMEDIATE 'DROP TABLE Persons CASCADE CONSTRAINTS';

EXCEPTION

WHEN OTHERS THEN NULL;

END;

/

BEGIN

EXECUTE IMMEDIATE 'DROP VIEW PersonDetails';

EXCEPTION

WHEN OTHERS THEN NULL;

END;

/

CREATE TABLE Persons (

PersonID NUMBER PRIMARY KEY,

FirstName VARCHAR2(50),

```
    LastName VARCHAR2(50),  
    Gender VARCHAR2(10),  
    DOB DATE,  
    Phone VARCHAR2(15),  
    Email VARCHAR2(100)  
);
```

```
CREATE TABLE Addresses (  
    AddressID NUMBER PRIMARY KEY,  
    PersonID NUMBER,  
    Street VARCHAR2(100),  
    City VARCHAR2(50),  
    State VARCHAR2(50),  
    Pincode VARCHAR2(10),  
    CONSTRAINT fk_person  
    FOREIGN KEY(PersonID)  
    REFERENCES Persons(PersonID)  
);
```

```
CREATE TABLE Education (  
    EducationID NUMBER PRIMARY KEY,  
    PersonID NUMBER,  
    Qualification VARCHAR2(100),  
    Institution VARCHAR2(100),  
    YearOfPassing NUMBER,  
    CONSTRAINT fk_edu_person  
    FOREIGN KEY(PersonID)  
    REFERENCES Persons(PersonID)  
);
```

```
CREATE TABLE Employment (  
    EmployeeID NUMBER PRIMARY KEY,  
    PersonID NUMBER,  
    CompanyName VARCHAR2(100),  
    Designation VARCHAR2(100),  
    Salary NUMBER(10,2),  
    CONSTRAINT fk_emp_person  
    FOREIGN KEY(PersonID)  
    REFERENCES Persons(PersonID)  
);
```

```
INSERT INTO Persons VALUES  
(1,'Arun','Kumar','Male',  
TO_DATE('15-05-2000','DD-MM-YYYY'),
```

```
'9876543210',  
'arun@gmail.com');
```

```
INSERT INTO Persons VALUES  
(2,'Divya','Rani','Female',  
TO_DATE('20-08-1999','DD-MM-YYYY'),  
'9876543211',  
'divya@gmail.com');
```

```
INSERT INTO Persons VALUES  
(3,'Rahul','Sharma','Male',  
TO_DATE('10-01-1998','DD-MM-YYYY'),  
'9876543212',  
'rahul@gmail.com');
```

```
INSERT INTO Addresses VALUES  
(1,1,'Anna Nagar','Chennai','Tamil Nadu','600040');
```

```
INSERT INTO Addresses VALUES  
(2,2,'RS Puram','Coimbatore','Tamil Nadu','641002');
```

```
INSERT INTO Addresses VALUES  
(3,3,'KK Nagar','Madurai','Tamil Nadu','625020');
```

```
INSERT INTO Education VALUES  
(1,1,'B.E CSE','Anna University',2021);
```

```
INSERT INTO Education VALUES  
(2,2,'B.Sc IT','Bharathiar University',2020);
```

```
INSERT INTO Education VALUES  
(3,3,'MCA','Madurai Kamaraj University',2022);
```

```
INSERT INTO Employment VALUES  
(1,1,'TCS','Software Engineer',45000);
```

```
INSERT INTO Employment VALUES  
(2,2,'Infosys','System Engineer',40000);
```

```
INSERT INTO Employment VALUES  
(3,3,'Wipro','Developer',42000);
```

```
COMMIT;
```

```
CREATE OR REPLACE VIEW PersonDetails AS  
SELECT
```

```
    p.PersonID,  
    p.FirstName,  
    p.LastName,  
    a.City,  
    e.Qualification,  
    em.CompanyName  
FROM Persons p  
JOIN Addresses a  
ON p.PersonID = a.PersonID  
JOIN Education e  
ON p.PersonID = e.PersonID  
JOIN Employment em  
ON p.PersonID = em.PersonID;
```

```
CREATE OR REPLACE PROCEDURE GetPersonInfo
```

```
(  
    p_id IN NUMBER  
)  
AS  
BEGIN  
    FOR rec IN  
    (  
        SELECT *  
        FROM Persons  
        WHERE PersonID = p_id  
    )  
    LOOP  
        DBMS_OUTPUT.PUT_LINE(  
            rec.FirstName || ' ' ||  
            rec.LastName  
        );  
    END LOOP;  
END;  
/
```

```
CREATE OR REPLACE FUNCTION GetSalary
```

```
(  
    p_id NUMBER  
)  
RETURN NUMBER  
IS  
    sal NUMBER;
```

```

BEGIN
    SELECT Salary
    INTO sal
    FROM Employment
    WHERE PersonID = p_id;

```

```

    RETURN sal;
END;
/

```

```

CREATE OR REPLACE TRIGGER Salary_Check
BEFORE INSERT OR UPDATE
ON Employment
FOR EACH ROW
BEGIN
    IF :NEW.Salary <= 0 THEN
        RAISE_APPLICATION_ERROR(
            -20001,
            'Salary must be greater than zero');
    END IF;
END;
/

```

```

SELECT * FROM Persons;

```

```

/*

```

OUTPUT:

PERSONID	FIRSTNAME	LASTNAME	GENDER	DOB	PHONE	EMAIL
1	Arun	Kumar	Male	15-MAY-00	9876543210	arun@gmail.com
2	Divya	Rani	Female	20-AUG-99	9876543211	divya@gmail.com
3	Rahul	Sharma	Male	10-JAN-98	9876543212	rahul@gmail.com

*/

```

SELECT * FROM Addresses;

```

```

/*

```

OUTPUT:

ADDRESSID	PERSONID	STREET	CITY	STATE	PINCODE
1	1	Anna Nagar	Chennai	Tamil Nadu	600040
2	2	RS Puram	Coimbatore	Tamil Nadu	641002

3 3 KK Nagar Madurai Tamil Nadu 625020
*/

SELECT * FROM Education;

/*

OUTPUT:

EDUCATIONID PERSONID QUALIFICATION INSTITUTION

1 1 B.E CSE Anna University
2 2 B.Sc IT Bharathiar University
3 3 MCA Madurai Kamaraj University

YEAROFPASSING

2021
2020
2022
*/

SELECT * FROM Employment;

/*

OUTPUT:

EMPLOYEEID PERSONID COMPANYNAME DESIGNATION SALARY

1 1 TCS Software Engineer 45000
2 2 Infosys System Engineer 40000
3 3 Wipro Developer 42000
*/

SELECT FirstName, LastName, Phone
FROM Persons;

/*

OUTPUT:

FIRSTNAME LASTNAME PHONE

Arun Kumar 9876543210
Divya Rani 9876543211
Rahul Sharma 9876543212

*/

```
SELECT FirstName, City
FROM Persons p
JOIN Addresses a
ON p.PersonID = a.PersonID;
```

/*

OUTPUT:

FIRSTNAME CITY

Arun	Chennai
Divya	Coimbatore
Rahul	Madurai

*/

```
SELECT FirstName, Qualification
FROM Persons p
JOIN Education e
ON p.PersonID = e.PersonID;
```

/*

OUTPUT:

FIRSTNAME QUALIFICATION

Arun	B.E CSE
Divya	B.Sc IT
Rahul	MCA

*/

```
SELECT FirstName, Salary
FROM Persons p
JOIN Employment e
ON p.PersonID = e.PersonID;
```

/*

OUTPUT:

FIRSTNAME SALARY

Arun	45000
Divya	40000

Rahul 42000

*/

```
SELECT GetSalary(1)
FROM Dual;
```

/*

OUTPUT:

GETSALARY(1)

45000

*/

```
SELECT * FROM PersonDetails;
```

/*

OUTPUT:

PERSONID	FIRSTNAME	LASTNAME	CITY	QUALIFICATION	COMPANYNAME
----------	-----------	----------	------	---------------	-------------

1	Arun	Kumar	Chennai	B.E CSE	TCS
2	Divya	Rani	Coimbatore	B.Sc IT	Infosys
3	Rahul	Sharma	Madurai	MCA	Wipro

*/

```
EXEC GetPersonInfo(1);
```

/*

OUTPUT:

Arun Kumar

*/

/*

TRIGGER TEST:

If salary is zero or negative, Oracle will display:

ORA-20001: Salary must be greater than zero

ORA-06512: at "SALARY_CHECK", line ...

Do NOT execute this test if you do not want an error message in the final output.

Example:

```
INSERT INTO Employment  
VALUES (4,1,'ABC Company','Tester',0);  
*/
```

```
SELECT * FROM Persons;
```

```
SELECT * FROM Addresses;
```

```
SELECT * FROM Education;
```

```
SELECT * FROM Employment;
```

```
SELECT * FROM PersonDetails;
```